

Faculty Vitae: Ying Liu

1. Education

B.S. Engineering Mechanics, Tsinghua University, 2001

M.A. Mechanical and Aerospace Engineering, Princeton University, 2004

Ph.D. Mechanical and Aerospace Engineering, Princeton University, 2007

2. Academic Experience

Department of Chemical Engineering, University of Illinois-Chicago

Assistant Professor, 2008-2015

Associate Professor, 2015-2021

Professor, 2021-2022

Dr. Satish C. & Asha Saxena Professor, 2022-current

Department of Biopharmaceutical Sciences, University of Illinois-Chicago

Adjunct professor, 2008-present

Department of Bioengineering, University of Illinois-Chicago

Adjunct professor, 2018-present

Department of Chemistry, University of Chicago

Post Doctoral Researcher, July 2007- July 2008

3. Non-Academic Experience

Not applicable.

4. Certifications or professional registrations

Not applicable.

5. Current membership in professional organizations

American Association for the Advancement of Science (AAAS)

American Institute of Chemical Engineers (AIChE)

American Physical Society (APS)

American Association of Pharmaceutical Science (AAPS)

Sino-American Pharmaceutical Association

6. Honors and awards

Sino-American Pharmaceutical Professionals Association Leadership Award, 2023

Chicago Biomedical Consortium, Catalyst Award, 2020

UIC Graduate Mentor Award, UIC, IL, 2017

UIC Researcher and Scholar of the Year, Rising Star Award, UIC, IL, 2016

CAREER Award, National Science Foundation, Jan. 2014 – Jan. 2019

College of Engineering Research Award, UIC, 2013

OTM POC Award, UIC, 2013

Chancellor Discovery Award, UIC, 2011

Technology for Developing Regions Fellowship, Princeton University, 2006

Howard Crathorne Phillips Graduate Fellowship, Princeton University, 2003

Engineering Fellowship, Princeton University, 2002
Rong Hong National Fellowship, Tsinghua University, 2000

7. Service activities

Interim Associate Dean of Research, 2024 – present
Engineering Dean Search Committee, 2023 – 2024
Director of Graduate Study, Department of Chemical Engineering, 2023 – 2024
UIC faculty senate, 2021 – 2024
College of Engineering, Executive Committee, UIC, 2016-2020, 2022 – present
Fellow of Honor's college, 2015 – present
Campus Research Board (CRB), 2016 – present
Member, Graduate Committee, Department of Chemical Engineering, University of Illinois at Chicago, 2008-2018
Student Sponsor for Illinois Math and Science Academy, 2010-2016
Mentor, Girls' Electronic Mentoring in Science, Engineering and Technology, 2009-present
Member, Editorial Board, Advances in Materials Science and Engineering, 2013-2019
Member, Editorial Board, Journal of Oncology and Biomarker Research, 2013-2018

8. Selected Publications Last 5 Years (Selected from 33 publications)

1. Han Xiao, Bailin Feng, Dongdong Gao, Shuangni Shi, Yiqing Yang, Yiwei Zhang, Fengyang Wang, Qi Yao, Haiqing Song, Ying Liu*, Gang Cheng*, A naturally derived lipopeptide lipid nanoparticle platform enabling multiple nucleic acids delivery, *Bioactive Materials*, 54, 829 – 849, 2025.
2. Bailin Feng, Shiyu Du, Yiqing Yang, Yuli Zhu, Nash McKeague, Binhua Lin, Wei Bu, Gang Cheng, Ying Liu*, Molecular Effects of Zwitterionic Peptide on Monolayer Lipid Membranes upon Enzyme-catalyzed Degradation, *Langmuir*, 41 (5), 3402-3412, 2025. PubMed PMID: 39895134.
3. Yuli Zhu; Lionel Porcar; Thirupathi Ravula; Krishna C. Batchu; Tera Lavoie; Ying Liu, Ursula Perez-Salas, Unexpected asymmetric distribution of cholesterol and phospholipids in equilibrium model membranes, *Biophysical Journal*, 123 (22), 3923-3934, 2024.
4. Kaige Ma+, Tiep Pham+, Jun Wang, InSug O-Sullivan, Amy DiCamillo, Shiyu Du, Fackson Mwale, Zeba Farooqui, Gina Votta-Velis, Benjamin Bruce, Andre J. van Wijnen, B. Duncan X. Lascelles, Ying Liu*, Hee-Jeong Im*, Nanoparticle-based Inhibition of Vascular Endothelial Growth Factor Receptors Resolves Osteoarthritis-Associated Pain and Cartilage Damage, *Sciences Advances*, 10 (7), eadi5501, 2024.
5. Alex Cheng, Ying Liu, and Haiqing Song, Elevating nucleic acid delivery via a stable anionic peptidedextran ternary system, *Biointerphases*, 18, 051001, 2023.
6. Hui Tang, Maryam Zaroudi, Yuli Zhu, Alex Cheng, Lei Qin, Bin Zhang, and Ying Liu*, Toroidal-Spiral Particles as A CAR-T Cell Delivery Device for Solid Tumor Immunotherapy, *Journal of Controlled Release*, 362, 620 – 630, 2023.
7. Chang Liu, Sherry Gong, and Ying Liu*, Micelle Morphology Phase Diagram in a Phospholipid, PEGylated Lipid, and Peptide Amphiphiles Ternary System, *Chemical Engineering Research and Design*, 181, 354-362, 2022.
8. Tiep Pham, Paola Leon Plata, Pin Zhang, Anand Vellara, Wei Bu, Binhua Lin, Gang Cheng, and Ying Liu*, Knowledge-based Design of 5-Fluorouracil Prodrug Liposomal Formulation:

Molecular Packing and Interaction Revealed by Interfacial Isotherms and X-ray Scattering Techniques, *Molecular Pharmaceutics*, 18, 12, 4331-4340, 2021.

9. Paola Leon Plata, Ludwig C. Nitsche, and Ying Liu*, Manipulation of Toroidal-Spiral Particles Internal Structure by Fluid Flow, *Physics of Fluids*, 33, 054102, 2021.

10. Paola Leon-Plata, Maryam Zaroudi, Chun-Yin Lee, Colin Foster, Ludwig C. Nitsche, Peter D. Rios, Yong Wang, Jose Oberholzer, and Ying Liu*, Heterogeneous Toroidal Spiral Particles for Islet Encapsulation, *Biomaterials Science*, 9, 3954, 2021.
511 (1), 415-423, 2016.

9. Professional Development

Not applicable